2.2 Measure Preliminaries-2

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Definition 0.1. Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. A mapping $f\colon X \to Y$ is called $(\mathcal{M}, \mathcal{N})$-measurable or measurable if $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{N}$.

Remark 0.1. The composition of measurable mappings is measurable.

Proposition 0.1. If $\mathcal{N}$ is generated by $\mathcal{E}$, then $f\colon X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable if and only if $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$.

Proof. The only if condition is trivial. For the if condition, observe that

$$\{E \subseteq Y \mid f^{-1}(E) \in \mathcal{M}\}$$

is a σ-algebra that contains $\mathcal{E}$ by hypothesis; it therefore contains $\mathcal{N}$. $\square$

Corollary 0.1. If X and Y are metric (or topological) spaces,

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Definition 0.2. Every continuous mapping is (B_X, B_Y)-measurable.

Definition 0.3. Let $(X, \mathcal{M})$ be a measurable space. A real or complex-valued function f on X is called **measurable** if f is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable (for real-valued f) or $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable (for complex-valued f).

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